

Question Paper

Exam Date & Time: 01-Jan-2026 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU

SEVENTH SEMESTER B.TECH. DEGREE,
MAKEUP EXAMINATIONS - JANUARY 2026

Deep Learning [IT_4442]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Evaluate the causes of overfitting and discuss the methods used to mitigate it in machine learning models. (5)
A)

Course Outcome: "CO1" / Cognitive Level: "Evaluate"

B) Evaluate the strengths and weaknesses of unsupervised learning algorithms in real-world applications such as customer segmentation or anomaly detection. (5)

Course Outcome: "CO2" / Cognitive Level: "Evaluate"
- 2) Evaluate the role of activation functions in deep feedforward networks. How does the choice of activation function affect the performance of the model? (5)
A)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

B) Evaluate the effect of different learning rates during the backpropagation process. How does an inappropriate learning rate impact model convergence? (5)

Course Outcome: "CO5" / Cognitive Level: "Evaluate"
- 3) Analyze the impact of data augmentation on the training time and model accuracy. Does augmentation always lead to better performance in deep learning models? (5)
A)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

B) Analyze the difference between early stopping and regularization in terms of their effect on model performance. In which situations would one be preferred over the other? (5)

Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Compare the effectiveness of gradient descent, stochastic gradient descent (SGD), and mini-batch gradient descent. How does each method affect convergence speed and stability? (5)
A)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

B) Analyze the architecture of a Convolutional Neural Network (CNN). How do convolutional layers and pooling layers contribute to the feature extraction process? (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) Compare the application of RNNs to other types of neural networks for tasks involving sequential data, such as time-series prediction or speech recognition. (5)

A)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Analyze the benefits of using LSTMs over traditional RNNs in applications such as natural language processing or speech recognition. How does LSTM address the vanishing gradient problem? (5)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 CSE_4064:Deep Learning

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	15
4	4	CO 4	5
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 CSE_4064:Deep Learning

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	0	0(0%)
4	Analysis	7	35(70%)
5	Create	3	15(30%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)